First Progress Review

Chapter 5: Work Plan and Timeline

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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5. Work Plan and Timeline

This chapter sets out the plan for the 36-month programme, which started on 1 October 2026 and ends on 30 September 2029. It reports where the work stands at the time of writing (10 November 2026, month 2), presents the Gantt chart, the detailed plan up to the First Progress Review (month 8, May 2027), milestones and deliverables, the publication plan, the supervision arrangements and the risk register. The plan keeps the structure of the Gantt chart in the accepted proposal. Two changes are made for clarity: the first progress review is shown as its own activity, and clinical validation (WP7), which the proposal describes in Year 3 but did not list in its chart, is added as a separate row.

5.1 Planning Principles

- **Front-load approvals and infrastructure.** Ethics approvals and the registry platform are started in Year 1 because every later work package depends on them.
- **Develop on public data while local data accrue.** WP2-WP4 start on public benchmarks (BraTS, TCGA) in Year 1, so that no work package waits for local data.
- **Core contribution in Year 2.** The escalation policy and orchestration (WP3) are developed and evaluated by the end of month 24.
- **Third-tier work last.** Federated learning (WP5) and the surgical-planning demonstrator (WP6) are scheduled last so that they can be narrowed without endangering completion.

- **Write as you go.** Each work package ends in a paper draft, so that thesis chapters are largely written before the final writing period.

5.2 Progress to Date

Table 5.1: Status of activities at 10 November 2026 (month 2).

Activity	Status	Evidence
PhD registration and proposal	Done	Proposal handed over to FGS on 1 October 2026; accepted by the Board and handed to evaluators on 4 November 2026
Literature collection	Done (first round)	Research Paper Bank of 559 open-access papers (O1 143, O2 109, O3 146, O4 161); 13 base papers; methodology extracted in detail for 246
Review tables	Done	One review sheet per objective, as instructed on 6 October 2026
Introduction (Chapter 1)	Draft complete	Draft version 1, 10 November 2026
Literature review (Chapter 2)	Draft complete	Draft version 1, 10 November 2026
Research methodology (Chapter 3)	Draft complete	Draft version 1, 10 November 2026, as instructed on 5 November 2026
Work plan and timeline (Chapter 5)	Draft complete	This chapter
KDU ethics application	Ready to submit	Signed by all investigators on 9 November 2026; REC fee paid on 10 November 2026 (receipt 13275); soft and hard copies prepared
USJ ethics application	Pending	To be submitted after KDU approval, as advised on 5 November 2026
Progress website	Done	Supervisors can follow updates, the log book, the timeline and the Paper Bank
Preliminary results (Chapter 4)	Not started	Public-data baselines planned for January-April 2027

5.3 Gantt Chart (36 Months)

Table 5.2: Gantt chart of the 36-month plan by quarter. Q1 = Oct-Dec 2026, Q2 = Jan-Mar 2027, Q3 = Apr-Jun 2027, Q4 = Jul-Sep 2027; Year 2 = Q5-Q8 (Oct 2027-Sep 2028); Year 3 = Q9-Q12 (Oct 2028-Sep 2029). Shaded cells show the quarters in which each activity runs.

No.	Activity	Months	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Status
1	**Literature review and proposal**														
1.1	Literature review, Paper Bank and proposal	1-8	■	■	■										In progress
1.2	Ethics approvals (KDU, USJ, hospitals)	2-12	■	■	■	■									In progress
1.3	First progress review report and presentation	2-8	■	■	■										In progress
2	**WP1: Registry platform and data pipeline**														
2.1	Registry, database and data architecture	5-20		■	■	■	■	■	■						Not started
2.2	Data ingestion and de-identification	9-24			■	■	■	■	■	■					Not started
2.3	Data-quality control; research data pipeline	17-24						■	■	■					Not started
3	**Data collection, annotation and modelling**														
3.1	WP2: MRI perception, calibrated uncertainty, local adaptation	5-24		■	■	■	■	■	■	■					Not started
4.0	WP3: Escalation policy and orchestration (core)	5-24		■	■	■	■	■	■	■					Not started
5.0	WP4: Pathology and combined diagnosis and staging	5-24		■	■	■	■	■	■	■					Not started
6	**Federation, surgical planning and validation**														
6.1	WP5: Federated learning across hospitals	21-36							■	■	■	■	■	■	Not started
7.0	WP6: Surgical-planning proof of concept	29-36										■	■	■	Not started
7.1	WP7: Clinical validation (multi-reader study)	25-34									■	■	■	■	Not started
8.0	Thesis writing and publication	29-36										■	■	■	Not started

Month ranges are counted from 1 October 2026 (month 1). Detailed month dates: month 8 = May 2027, month 12 = September 2027, month 20 = May 2028, month 24 = September 2028, month 29 = February 2029, month 36 = September 2029.

5.4 Detailed Plan to the First Progress Review

Table 5.3 gives the month-by-month plan from now until the First Progress Review, due on 10 May 2027.

Table 5.3: Monthly plan, November 2026 to May 2027 (months 2-8).

Month	Planned work	Output
Nov 2026 (M2)	Finalise Introduction, Literature Review, Methodology and Work Plan drafts; supervisor review; submit and follow up the KDU ethics application; continue the Paper Bank (fill remaining methodology extractions)	Chapters 1, 2, 3, 5 version 1; KDU ethics submitted
Dec 2026 (M3)	Revise chapters after supervisor comments; set up the development environment (PyTorch, MONAI, nnU-Net); download and pre-process BraTS 2021 and BraTS-Africa; elicit draft error costs with the clinical supervisor	Chapters version 2; reproducible pipeline
Jan 2027 (M4)	WP2 baselines: nnU-Net and Swin UNETR on BraTS; Monte Carlo dropout, ensembles and temperature scaling; respond to ethics committee queries	Baseline Dice and calibration results
Feb 2027 (M5)	WP3 prototype: orchestrator with MRI and classification agents on public data; first risk-coverage curves for the escalation rule versus fixed thresholds; registry requirements and data model (WP1)	Prototype; first escalation results
Mar 2027 (M6)	WP4 start: CLAM with a pathology foundation encoder on TCGA glioma slides; registry prototype (FHIR model, Orthanc); USJ ethics application after KDU approval	Pathology baseline; registry prototype
Apr 2027 (M7)	Write Chapter 4 (Preliminary Work and Results); assemble the full report; draft a review paper from Chapter 2	Chapter 4; full report draft
May 2027 (M8)	Supervisor sign-off; submit the First Progress Review report (10 May 2027); prepare and deliver the presentation	Report and presentation submitted

5.5 Milestones and Deliverables

Table 5.4: Milestones.

No.	Milestone	Month	Target date
M1	Proposal accepted; registration approved	2	Nov 2026 (achieved 4 Nov 2026)
M2	KDU ethics application submitted	2	Nov 2026 (fee paid 10 Nov 2026)
M3	KDU and USJ ethics approval	6-12	by Sep 2027
M4	First Progress Review report submitted	8	10 May 2027
M5	Public-data baselines for MRI, pathology and escalation (WP2-WP4)	8	May 2027
M6	Registry platform operational at the first site (WP1)	12-20	by May 2028
M7	Local MRI models calibrated and evaluated (WP2)	20	May 2028
M8	Escalation policy and orchestration evaluated (WP3, core contribution)	24	Sep 2028
M9	Combined diagnosis and staging evaluated (WP4)	24	Sep 2028
M10	Federated training across hospitals evaluated (WP5)	33	Jun 2029
M11	Clinical validation completed (WP7)	34	Jul 2029
M12	Surgical-planning proof of concept evaluated (WP6)	35	Aug 2029
M13	Thesis submitted	36	Sep 2029

Table 5.5: Deliverables by work package.

WP	Deliverable	Due (month)
WP1	National Brain Tumour Patient Registry with de-identification, quality control and audit trail	20

WP	Deliverable	Due (month)
WP2	Calibrated, data-efficient MRI segmentation and classification models; failure taxonomy	20
WP3	Orchestrator and escalation policy; risk-coverage and workload-safety evaluation	24
WP4	Pathology agent and fusion model robust to missing sources	24
WP5	Federated training protocol and calibration-after-aggregation analysis	33
WP6	Surgical-planning demonstrator with structures-at-risk map	35
WP7	Multi-reader clinical evaluation and interview study; escalation-quality protocol	34
All	De-identified Sri Lankan brain-tumour benchmark (subject to approvals); thesis	36

5.6 Publication Plan

Table 5.6: Planned publications. Target venues are indicative and will be agreed with the supervisors.

No.	Paper	Work package	Target submission	Indicative venue
P0	Review: uncertainty-aware, human-in-the-loop multi-agent AI in clinical imaging (from Chapter 2)	Review	Month 10 (Jul 2027)	npj Digital Medicine / Artificial Intelligence in Medicine
P1	Data-efficient, calibrated brain-tumour MRI models on Sri Lankan data	WP2	Month 20 (May 2028)	MICCAI / Medical Image Analysis
P2	Calibrated selective autonomy for a multi-agent clinical pathway (flagship)	WP3	Month 24 (Sep 2028)	Nature Communications / npj Digital Medicine
P3	Pathology and MRI fusion for WHO 2021 diagnosis with missing sources	WP4	Month 28 (Jan 2029)	IEEE Transactions on Medical Imaging
P4	Federated learning across Sri Lankan hospitals and its effect on calibration	WP5	Month 33 (Jun 2029)	Radiology: Artificial Intelligence
P5	Clinical evaluation of the human-in-the-loop framework (optional)	WP7	Month 35 (Aug 2029)	The Lancet Digital Health / JAMIA
P6	Surgical-planning proof of concept (optional)	WP6	Month 36 (Sep 2029)	Neurosurgical Focus / IJCARS

5.7 Supervision and Reporting

- Meetings with the principal supervisor (Dr. WMKS Ilmini) every two weeks, and with the co-supervisor (Dr. WGCW Kumara) monthly, on research progress and technical decisions.
- Monthly meetings with the clinical co-supervisor (Dr. AM Deepal K Attanayake) on data access, error costs, annotation and clinical evaluation.
- Every meeting is recorded in the log book on the progress website with decisions and action items; updates, the timeline and the Paper Bank are kept current there.
- Progress reviews to the Faculty of Graduate Studies as scheduled by the Faculty, starting with the First Progress Review in May 2027.

5.8 Risk Register

Table 5.7: Main risks and mitigations.

Risk	Likelihood	Impact	Mitigation
Ethics or hospital approvals delayed	Medium	High	Applications started early (KDU finalised and fee paid on 10 Nov 2026); development continues on public data; approvals tracked monthly

Risk	Likelihood	Impact	Mitigation
Fewer usable local cases than expected	Medium	High	Data-efficient methods (transfer, self-supervised); public pre-training; federated learning; evaluation designed around hundreds, not thousands, of cases
Biopsy slides cannot be digitised at scale	Medium	Medium	Use slides already scanned; prioritise a subset; WP4 develops on TCGA; fusion handles missing tissue
Clinician time for annotation and reading is scarce	High	Medium	Multi-reader, multi-case design; model-assisted annotation; short sessions; double annotation only for a subset
Only one or two hospitals join the federation	Medium	Medium	Simulated federation by site partition; narrow WP5 (third tier) without affecting the core
Computing resources insufficient	Medium	Medium	University GPU resources; cloud credits for research; efficient models (nnU-Net configurations, frozen encoders)
Calibration does not hold under shift	Medium	High	Multi-domain temperature scaling, ensembles, per-site recalibration and re-checking after every update; this is also a research finding
Scope creep	Medium	Medium	Tiered scope; surgical planning fixed as proof of concept; scope reviewed at each progress review